

BASF
We create chemistry

Care Creations™

Ultra Filling Spheres™

Plump up the volume

by Beauty Creations
another Care Creations™ product group Inspired by Life

Global Except US

TABLE OF CONTENTS

Summary file 3

Ultra Filling Spheres, extreme plumping 4

Instant anti-wrinkle effect, an overview 5

Reaching new spheres 6

Demonstrated efficacy 8

Observation of the swelling power of the spheres by optical microscopy (*in vitro*) 9

Demonstration of the swelling power of the spheres by an *in vitro* test (*in vitro*) 10

Water absorption capacity by dynamic vapor sorption measurement (*in vitro*) 11

Stratum corneum swelling test (*ex vivo*) 12

★ **New test** - Moisturizing activity in the model of human *stratum corneum* - Tagami test (*in vivo*) 13

★ **New test** - Volumizing efficacy for the lips (*in vivo*) 14

★ **New test** - Smoothing efficacy for the face (*in vivo*) 15

General conclusion 16

Annexes 17

Technical data 17

Clinical test formula 17

Formulation examples 18

Bibliography 20

The trademarks symbolized with a ® or ™ are either property of or licensed to BASF group and registered and/or applied for registration in relevant countries. Other product names and trademarks mentioned may belong to third parties.

SUMMARY FILE

Ultra Filling Spheres®	C00487
Origin - Description	Dehydrated microspheres of hyaluronic acid and konjac glucomannan, suspended in an oily medium of ethylhexyl palmitate
Regulatory data	
INCI	Ethylhexyl Palmitate (and) Trihydroxystearin (and) Sodium Hyaluronate (and) Glucomannan
China	Each component is listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2021)
CAS#	29806-73-3, 139-44-6, 9067-32-7, 37220-17-0
EINECS#	249-862-1, 205-364-6, 232-678-0, 253-404-6
Appearance	Off-white opaque syrupy liquid, with characteristic odor
Preservative	None
Halal	Yes
Cosmetic use	
Properties	Wrinkle smoothing Volumizing effect Long lasting hydration
Applications	Anti-wrinkle treatment Eye contour, mature skin Deep wrinkle treatment (frown lines, forehead wrinkles, nasolabial folds) Deep hydration treatment Fast lip volume replumper
Formulation data	
Concentration of use	0.5 - 2%
Solubility	Soluble in oil
Incorporation method	Ultra Filling Spheres is incorporated into the oily phase under low stirring
Optimal temperature of use	15 - 95°C
Optimal pH	4 - 8



ULTRA FILLING SPHERES, EXTREME PLUMPING

The groundbreaking association of low molecular weight hyaluronic acid with konjac glucomannan results in the creation of the most effective Filling Spheres ever seen.

Back in the early 2000s, BASF Beauty Creations launched a revolutionary anti-aging technology on the market: Filling Spheres.
Even at that time, these dehydrated microspheres of vegetable or marine origin were already capable of smoothing wrinkles in just a few hours, thanks to their small size, their ability to capture water and their plumping action.

Inspired by the wrinkle fillers used in aesthetic medicine, the Hyaluronic Filling Spheres (HAFS) were launched in 2005.

Fast forward to the present - just about everyone knows Hyaluronic Filling Spheres. They’re a technological, scientific and marketing success and a great classic among cosmetic ingredients for an instant anti-wrinkle effect, just like tensors.

In a constant quest for innovation and effectiveness, BASF Beauty Creations has created a fourth generation of Filling Spheres - Ultra Filling Spheres (UFS) - with even greater plumping power.
Through the association of konjac, and low molecular weight hyaluronic acid, Ultra Filling Spheres capture 20% more water and plump up 4 times better than Spheres of the previous generation!

INSTANT ANTI-WRINKLE EFFECT, AN OVERVIEW

The instant anti-wrinkle effect is an essential component of anti-aging cosmetic products. Four main approaches are in competition: tensors, pigments and Filling Spheres.

Tensors

With their film-forming properties and their ability to retract after application, cosmetic tensors produce a visible and perceptible effect. Users feel a tightening of the skin, micro protrusions and wrinkles appear to be smoothed. Tensors can be classified into two major families, according to their origin; proteinic or polysaccharidic.

The historical reference on the subject is a protein, albumin. Back in 1965, Albert Kligman already described the anti-wrinkle effects of albumin. His conclusion was simple: the anti-wrinkle effect of albumin is due to the formation of a film that raises the furrows of the skin and reduces wrinkles on the surrounding skin (Kligman AM *et al*, 1965).

Animal proteins, since then fallen from grace, were replaced by vegetal proteins, often less effective. Most tensors used today are cross-linked or hydrolyzed polysaccharides.

Almost all present the same drawbacks: no long-term effect, elimination in contact with water, rapid loss of the effects in case of repetitive movements and high use concentrations required (4-10%). Some film-forming and waterproof silicones have similar applications, but the tightening effect remains much lower.

Pigments

"Blurred effect pigments", better known as "soft-focus" pigments, fill the wrinkle to diffuse light evenly and lighten dark zones. Of course, it's only an optical, "trompe-l'oeil" effect, however well done. Still, the use of these pigments is mainly reserved for makeup, and is usually unsuitable for care products.

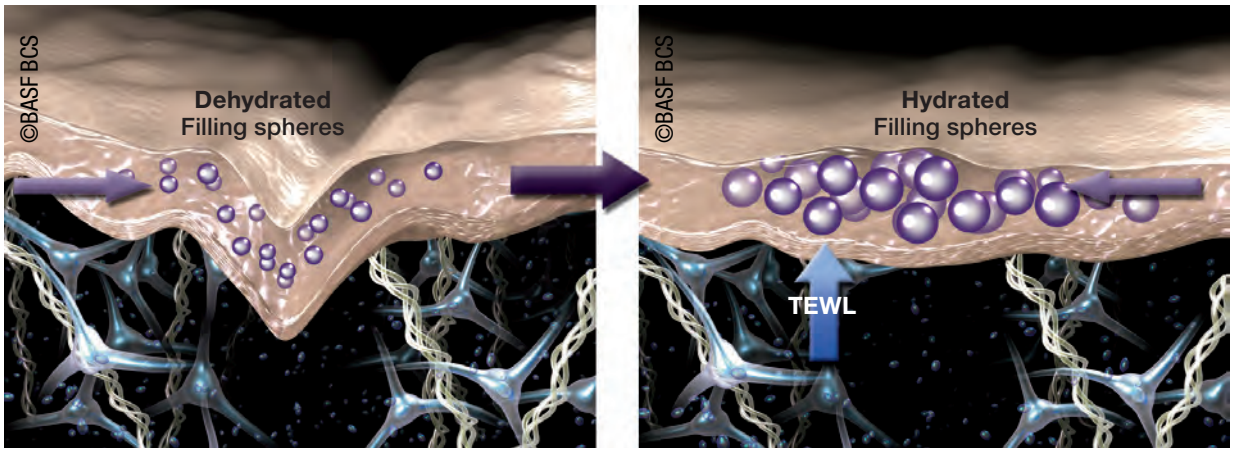
Filling Spheres

Inspired by injectable wrinkle fillers and developed through the unique expertise of BASF Beauty Creations in the field of vectorization, the Filling Spheres are now one of the essential cosmetic ingredients.

These microspheres are based on vegetal polysaccharides, marine collagen or hyaluronic acid, and first penetrate the upper layers of the epidermis before capturing the water that evaporates naturally, plumping and smoothening the wrinkle from within.

The specificity of the market creates a permanent need for innovation and rapid product turnover to meet consumer expectations; Filling Spheres are no exception to this rule.

Taking advantage of its experience in this field and its capacity to innovate, Beauty Creations is launching a fourth generation of Filling Spheres, Ultra Filling Spheres.



REACHING NEW SPHERES

To go beyond the hygroscopic power of hyaluronic acid is a real challenge, but that's just what Ultra Filling Spheres.

1.000 times its own weight. That's the amount of water that hyaluronic acid can absorb. This is the origin of the power of hyaluronic acid. But can we do better? As the saying goes «They didn't know it was impossible, so they did it»…
To develop its next generation of Filling Spheres, BASF Beauty Creations chose to look for the possible synergies between hyaluronic acid and biocompatible, biodegradable natural polymers, since biocompatibility and biodegradability are trademarks of Filling Spheres.
Arabinans, galactans, mannans, arabinogalactans, galactomannans: all kinds of gums, there were plenty of candidates. However, one contender proved particularly effective for the desired application: konjac glucomannan.

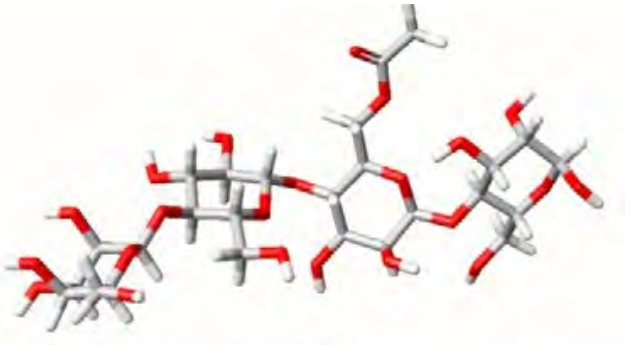
Konjac, the slimming sugar

Konjac -the generic term designating the different products obtained from the bulb of Amorphophallus konjac- has been used for more than 2,000 years in traditional Chinese medicine to treat asthma, cough, and various skin diseases. Introduced in Japan during the 7th century, it's there that it really took off.
Today, konjac flour is an essential ingredient in Japanese cuisine (Chuaa M *et al*, 2010). And the reason why konjac has gained a lot of publicity recently is because of its nutritional benefits. The main component of konjac is konjac glucomannan. Konjac glucomannan is a dietary fiber that's not hydrolyzed by digestive enzymes, contains no calories, and swells to reach up to 200 times its own volume. It is thus an ideal appetite suppressant (Arvill A *et al*, 1995).
But there's more about konjac. It also modifies the absorption kinetics of other nutrients. Used as a dietary supplement, it slows down fat and sugar absorption and helps lower cholesterol and glucose serum levels (Vasques Carlos AR *et al*, 2008).
Today, we find konjac glucomannan in most slimming diets as basic ingredient or as a complement. We are progressively discovering other merits of konjac: it is believed to have anti-senescence (Lu XJ *et al*, 2002) and probiotic (Al-Ghazzewi FH *and Tester*, 2010) properties.
In the field of cosmetics, konjac glucomannan is known for its capacity to act on skin regeneration (Heber G, 2008) and for its use as wrinkle-filling substance (Heber G, 2008).
Glucomannan gels also present interesting properties for the inclusion of cells required for cytopathological studies (Kaneko C *et al*, 2010), tissue engineering (Kondo T *et al*, 2009) and programmed release of small molecules (Alvarez-Manceñido F *et al*, 2008).


Figure 1 - Konjac plant.

Ultra Filling Spheres

The first generations of Filling Spheres were obtained from macromolecules of marine origin (atelocollagen, chondroitin sulfate) or vegetal proteins (wheat, carob, acacia). The use of hyaluronic acid, and even more so of low molecular weight hyaluronic acid, made it possible to increase the capacity to re-inflate the sphere even more and to achieve real anti-wrinkle effects. Today, thanks to the association of low molecular weight hyaluronic acid with Konjac glucomannan, Ultra Filling Spheres take the re-inflation capacities of the Filling Spheres to new levels. Effectiveness tests prove it: compared to Hyaluronic Filling Spheres, Ultra Filling Spheres capture more water (1.4 times more), capture it faster, and once they are rehydrated, fill up a volume that's 4 times greater!


Figure 2 - Repetition pattern of konjac glucomannan (glucose/mannose).

More, faster and with less water: the perfect dosage of low molecular weight hyaluronic acid and konjac glucomannan improves the re-inflation capacity of the Ultra Filling Spheres for ever-increasing effectiveness in record time.

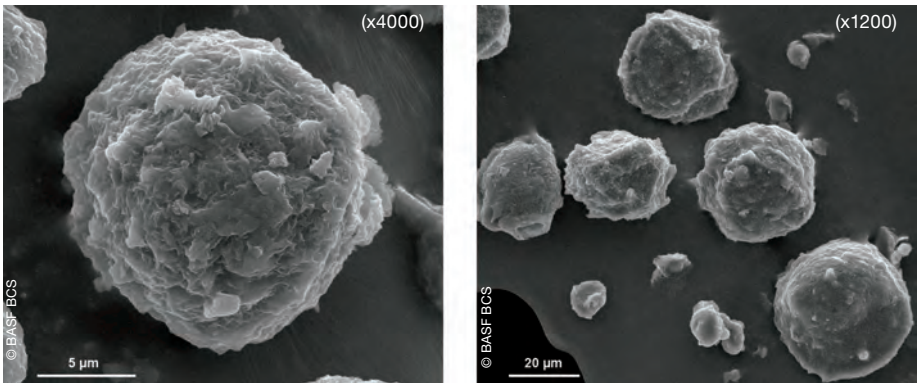


Figure 3 - Scanning electron microscopy pictures of dehydrated Ultra Filling Spheres (Quanta 250 FEG, FEI, USA).

Safety/Tolerability

UFS was tested to ensure its safety under the recommended conditions of use.

Description

Ultra Filling Spheres are dehydrated microspheres of hyaluronic acid and konjac glucomannan, suspended in an oily medium of ethylhexyl palmitate.

INCI name: Ethylhexyl Palmitate (and) Trihydroxystearin (and) Sodium Hyaluronate (and) Glucomannan

Naturalness content (according ISO 16128): 70.6% from natural origin

Recommended dose of use: 0.5 - 2%

China IECIC listed

Vegan Friendly **



*Example: illustration of a commercial sample of Ultra Filling Spheres in: **A** - W/O emulsion with 2% UFS, **B** - Commercial product UFS, **C** - Lipstick with 2% UFS, **D** - O/W emulsion with 2% UFS.*

**No involvement of animal genes or animal-derived substances; unlikely cross-contamination from animal substances; No animal testing has been carried out by or on behalf of BASF on the ingredients of the product after 11th March 2009 and/or after the extended deadline 11th March 2013 for the purposes of the Cosmetic Regulation (EU)1223/2009 (statement available upon request).



DEMONSTRATED EFFICACY

The efficacy of this new generation of Filling Spheres was proven by *in vitro* and *in vivo* tests.

***In vitro* performance**:

- Swelling of the spheres observed by optical microscopy.
- Swelling of the spheres demonstrated by an *in vitro* test.
- Water absorption capacity by measurement of dynamic vapor sorption.
- Swelling of the corneous layer through an *ex vivo* test on skin biopsies.

★ **New test** - Moisturizing activity in the model of human *stratum corneum.*

***In vivo* performance**:

★ **New tests** - Ultra Filling Spheres (UFS) has been shown to have smoothing efficacy on the face and volumizing efficacy on the lips.

EFFICACY

Observation of the swelling power of the spheres by optical microscopy

in vitro

OBJECTIVE

The goal of this study is to evaluate the swelling power of the Ultra Filling Spheres (UFS) by optical microscopy. The images are analyzed to calculate the area of the dehydrated spheres in comparison with the area of the rehydrated spheres.
The swelling factor through this technique is defined as the ratio between these 2 areas.

RESULTS & DISCUSSION

When the UFS were observed in their dried form with an optical microscope, they appeared small and dense. After rehydration, they appeared larger and very round.

The areas of each form of spheres were analyzed by image processing software (Figure 4).

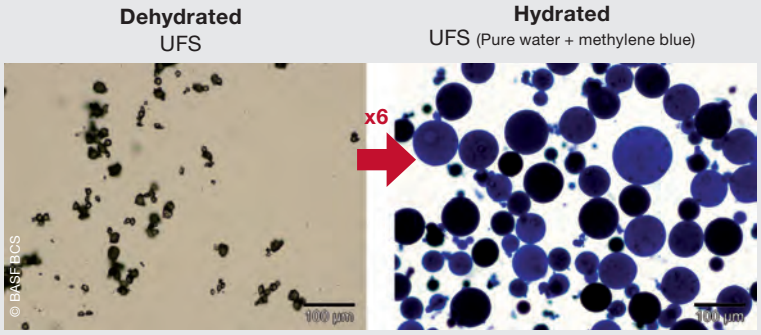


Figure 4 - Optical microscope pictures before and after the swelling of the UFS.

CONCLUSION

The optical microscopy technique confirms the exceptional swelling behavior of the UFS, when in contact with water.

MATERIALS&METHODS

Tested product
UFS in powder form (dehydrated spheres).

Protocol
UFS were observed in their dehydrated and rehydrated form (after 10 minutes in demineralized water). In the rehydrated form, the spheres were stained with methylene blue before microscopic observation.

Optical microscopy
The observations were made with an optical microscope Axioskop 2+ (Zeiss, Germany).
The photos were taken with a digital camera Dwm1200f (Nikon, France) and analyzed by the software NIS-Elements (Nikon, France).

Area measurement
UFS are similar to a sphere whose surface is $\pi * r^2$.
The area is calculated and estimated in pixels.

Results
The swelling factor is defined as the ratio of the average of the area of the rehydrated spheres / average of the area of the dehydrated spheres.

EFFICACY

Demonstration of the swelling power of the spheres by an *in vitro* test

in vitro

OBJECTIVE

The goal of the study is to quantify the swelling power of UFS *in vitro,* by measuring two parameters:
The quantity of water absorbed by the dehydrated sphere powder.
The swelling rate or the ratio of the volumes of the hydrated spheres over the dehydrated spheres.

RESULTS & DISCUSSION

Quantity of water absorbed by the dehydrated spheres

After 1 hour of contact with pure water, the swelling of the dried spheres (UFS or HAFS) in the tube was at its maximum.
After evaluation of the quantity of water caught by the Filling Spheres, this experiment showed that the UFS can absorb up to **25 times** their weight in water (Figure 5).
This water absorption capacity is significantly greater than that of the HAFS by a factor of **x1.4**.
Since the water reserve is reinforced, we can assume that there is a greater persistence of hydration of the upper layers of the epidermis.

	Water absorbed by the dehydrated spheres (quantity in g)	Water absorption capacity (mass ratio rehydrated/dehydrated)
HAFS	3.49 ± 0.15	17.5
UFS	4.93 ± 0.23	24.7**

x 1.4

Figure 5 - Measurement of water absorption parameters for the Filling Spheres.
Average ± SD, n=6.
*** statistically significant vs HAFS, p<0.01.*

Swelling rate

The measurement of the height of the gel formed by the rehydrated Filling Spheres, shows that the UFS can swell up to 16 times their initial volume (Figure 6).

The UFS have a swelling capacity 4 times higher than the HAFS.

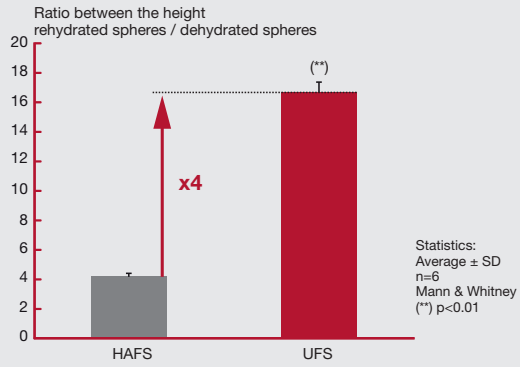


Figure 6 - Swelling rate of Filling Spheres.

Illustrations of the *in vitro* test

For each type of spheres (Figure 7):
- Left tubes: dehydrated Filling Spheres.
- Right tubes: rehydrated Filling Spheres.

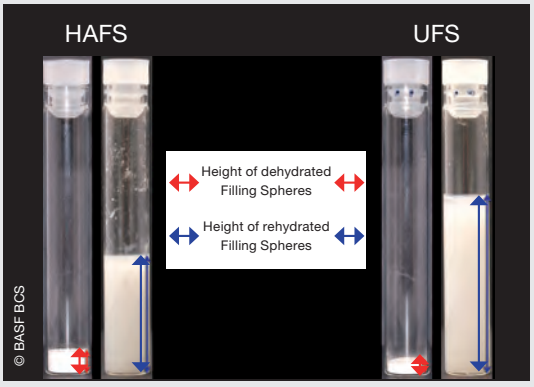


Figure 7 - Swelling of the spheres *in vitro*.

CONCLUSION

The swelling rate of UFS is higher than those found for HAFS. Consequently, UFS is able to absorb more water than HAFS.

MATERIALS&METHODS

Tested products

UFS and HAFS in powder form (dehydrated spheres).

Protocol

0.2 g of dehydrated Filling Spheres were put in a flat-bottom test tube, previously weighed (tare weight).
The powder was then compacted according to a standardized method and the height (Hdry) was measured with a graduated ruler.
10 ml of distilled water was inserted into the tubes with a syringe, in two phases. After the introduction of the 1st half (5 ml), the tube was placed under strong agitation (Vortex).
The 2nd half of the water (5 ml) was added up to the top of the tube. Once closed, this tube was placed under Vortex agitation then left to rest for approximately 1 h.
In this condition, The Filling Spheres reached their maximum swelling.
The tube was once placed under Vortex agitation then centrifuged at 4000 rpm for 4 min. Excess water was eliminated and the tube was then weighted (Mtot). The amount of water was determined as follows:
Qwater = Mtot – Tare weight – 0.2.
In the meantime, the height of the gel formed by the rehydrated spheres (Hwet) was measured using a graduated ruler.

Results

Water absorption capacity is defined as the ratio between the weight of the rehydrated spheres / weight of the dehydrated spheres (Qwater / 0.2)
The swelling rate is defined as the ratio of the height of the rehydrated spheres / height of the dehydrated spheres. (Hwet/Hdry).

Statistics

The test was performed in n=6.
The difference between the HAFS and UFS products was evaluated using the Rank Sum Test of Mann & Whitney (significance **: p< 0.01).

EFFICACY

Water absorption capacity by dynamic vapor sorption measurement

in vitro

OBJECTIVE

The objective of this study is to quantify the *in vitro* water retention of Filling Spheres by monitoring the variation of the spheres mass over time, at different levels of relative humidity between 0 and 90%.
In this study, UFS are compared to Hyaluronic Filling Spheres (HAFS), in terms of water absorption capacity.

RESULTS & DISCUSSION

Dynamic Vapor Sorption (commonly called DVS) is a gravimetric technique used to study the interactions of vapors with solids.
The dynamic process of water sorption by powders can be considered as a diffusion process in which the free water present in the air / powder boundary layer penetrates to a depth ranging from low to high depending on the sensitivity to humidity of the powder (absorption).
The dynamic vapor sorption curves of UFS and HAFS were established as shown on Figure 8.
The red curve represents the sorption isotherm of UFS, while the grey one represents the sorption isotherm of HAFS. Both powders show a high water sorption capacity. They absorb water quickly, especially when humidity levels are very high. At the maximum absorption level, i.e. at a relative humidity level of 90%, UFS increase their weight by 54% while the weight of HAFS went up by 45%. According to this method, the UFS thus retains 20% more water than the HAFS.
During desorption phase, which can be assimilated to a drying step, mass variation decreases. Humidity absorption is globally reversible for both powders which indicates that this absorption does not modify the structure of the materials.
In addition, we can note that UFS release humidity more slowly than the HAFS, showing that UFS have higher water retention power.

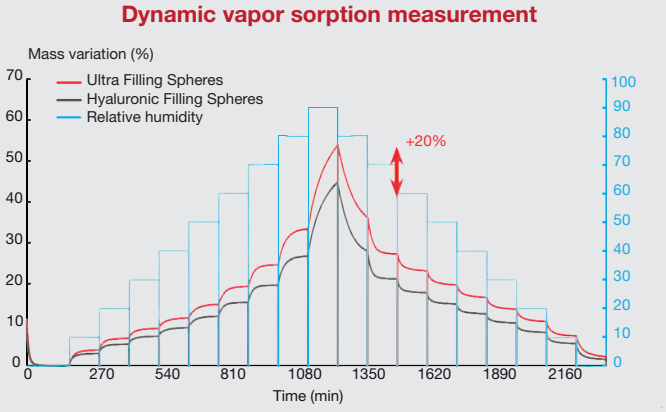


Figure 8 - Dynamic Vapor Sorption profile of UFS and HAFS.

CONCLUSION

The dynamic vapor sorption (DVS) measurement technique is perfectly adapted to quantify the swelling and water retention capability of Filling Spheres. Using this technique, we demonstrated that UFS is able to better capture water but also to retain it, compared to HAFS. Thanks to their strong hygroscopic power, UFS have faster and longer absorption power than HAFS.

MATERIALS&METHODS

Tested products

UFS and HAFS in powder form (dehydrated spheres).

Principle

By varying the concentration of vapor, or relative humidity (RH) surrounding the sample and by measuring the resulting variation of mass, the speed and the quantity of water absorbed by the sample can be measured.
The dynamic vapor sorption (*DVS* equipment (Figures 9 and 10) consists of a microbalance placed in a thermostatically controlled enclosure, measuring the absorption and desorption of water vapor from the sample. Measurements are usually performed at constant pressure and with a fixed total flow, in order to minimize mass variations. The use of a fixed temperature provides the sorption isotherm curve through the equilibration of the sample at a series of vapor concentrations.



Figure 9 - Illustration of the apparatus DVS Q5000 SA. Source: TA Instruments.

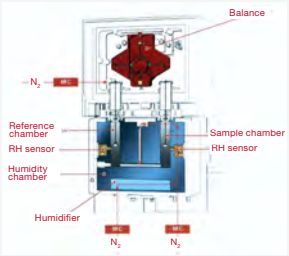


Figure 10 - Scheme of the apparatus DVS Q5000.

Protocol

Trials were performed with the DVS Q5000 SA by TA Instruments.
After drying at a relative humidity (RH) level of zero (0% RH) for 120 minutes, humidity was injected in increments from 10% RH to 90% RH.
The protocol parameters can be summarized as follows:

- nitrogen atmosphere 200 ml/min,
- pre-drying at 60°C under 0% RH 120 min,
- isotherm: 25°C for 120 min for each level,
- moisture levels of 10 to 90% RH and from 90% to 10% RH,
- metallic quartz crucibles,
- sample mass of about 5 mg of powder for all tests.

The change in mass of the sample is measured until equilibrium is reached.

EFFICACY

Stratum corneum swelling test

ex vivo

OBJECTIVE

The objective of this study, done on human biopsies is to show the swelling power of the UFS in comparison with the HAFS by measuring the *stratum corneum* thickness. A new visualization method, as well as a quantitative evaluation of *stratum corneum* thickness variations were used in this study.

RESULTS & DISCUSSION

After 4 hours, the thickness of the *stratum corneum* treated with UFS is higher than those treated with HAFS (+17%, $p<0.05$).

UFS showed a swelling power of the *stratum corneum* significantly higher than HAFS (Figure 11 and 12), proving that they can absorb more water than HAFS.

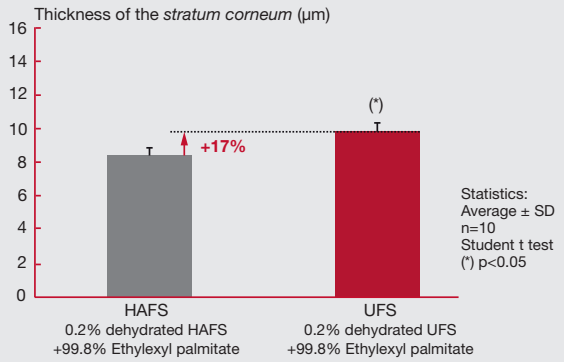


Figure 11 - *Ex vivo* test on the swelling of the *stratum corneum* by Filling Spheres after 4 hours.

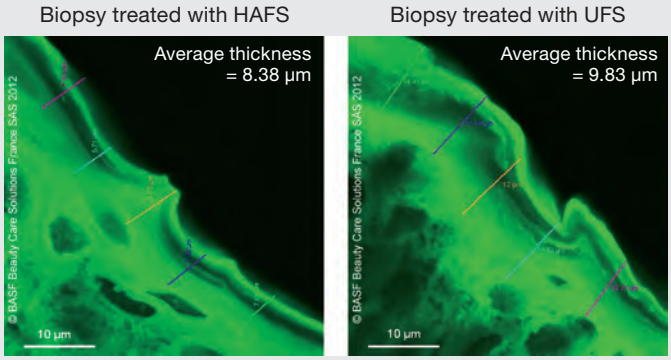


Figure 12 - Illustration of the quantification of the *stratum corneum* thickness on human biopsies.

CONCLUSION

After 4 hours, the swelling of the *stratum corneum* treated with UFS is significantly higher to those treated with HAFS.

MATERIALS&METHODS

Tested products

UFS and HAFS, commercial products (spheres diluted into ethylhexyl palmitate).

Principle

The thickness of the *stratum corneum* was evaluated, consisting of approximately 15 cell layers was evaluated after skin treatment with UFS and HAFS.

Protocol

The treatment of human skin biopsies, was carried out under controlled atmosphere at 44% relative humidity.

After balancing the biopsies for 16h, the products to be tested (UFS or HAFS) were applied for 4 hours at a dose of 4 mg/cm². At the end of the treatment, biopsies were frozen and kept in liquid nitrogen.

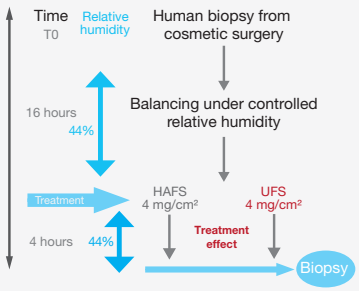


Figure 13 - Biopsy treatment protocol for the the *ex vivo* test on the swelling of the *stratum corneum*.

10 µm sections were formed, fixed in acetone, contrasted with the FITC fluorophore (0.1% solution), rinsed and then mounted in a glycerol buffer.

Confocal microscopy

Observations and analyzes were performed using a TCS-SP2 confocal microscope (Leica, Germany).

The samples were observed after excitation at 488 nm (corresponding to the excitation wavelength of FITC) and image acquisition was done between 510 and 530 nm. The thickness of the *stratum corneum* was then measured.

Results & statistics

The thickness of the *stratum corneum* was evaluated in 10 biopsies sections (5 measurements per slices). Statistical analysis was done using Student t test. The threshold of significance was set to 5% ($p<0.05$).

Statistics

The test was performed in n=10 (10 sections of 5 measurements per biopsy) the difference between the products UFS and HAFS was assessed by Student t test (significance * $p<0.05$) on the averages obtained for the measurement of the thickness of the *stratum corneum*.

EFFICACY

★ New test - Moisturizing activity on human *stratum corneum* (Tagami test)

in vitro

OBJECTIVE

The purpose of this study was to demonstrate the hydrating power of the UFS.
The dielectric conductivity of human skin biopsies treated with 2% UFS formulated in a hydrogel was compared with biopsies treated with the hydrogel alone.

RESULTS & DISCUSSION

The moisturizing activity of UFS at 2% in a hydrogel was assessed by the measurement of the dielectric conductivity (Figure 14).

Inherently weakly hydrated, the *stratum corneum* has a weak electric conduction. Once hydrated, the *stratum corneum* is more sensitive to the electric field, because of the bipolarity of water molecules. Thus, the application of topical moisturizing agents causes an increase in the conductance values: more the *stratum corneum* is moisturized, the higher the conductance value is.

A hydrogel vehicle (called "placebo hydrogel") was evaluated because of its properties to increase the state of hydration of the *stratum corneum* and therefore the dielectric conductivity.

The results showed that UFS at 2% increases the dielectric conductivity in the *stratum corneum* significantly compared to the placebo hydrogel and this from 30 min up to 6 hours after application.

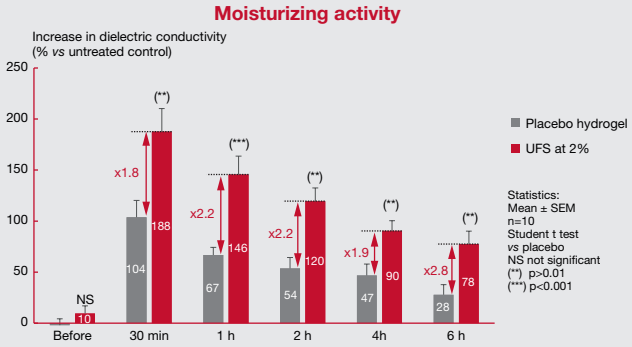


Figure 14 - Moisturizing power of UFS.

CONCLUSION

Once swollen on the epidermis, UFS improved immediately and with a long-lasting effect the skin moisturization.

MATERIALS&METHODS

Material

The Hygrometer Skicon 200 (IBS Inc., Hamamatsu, Japan) was used to measure the conductance in µS (microSiemens).

Treatment

This study was conducted with *stratum corneum* isolated from human skin explants from plastic surgery. They were prepared by peeling of the dermis after heating at 60°C for 2 minutes. The *stratum corneum* was then isolated by enzymatic digestion as described in the publication (Rochefort A *et al*, 1986).

Isolated *stratum corneum* were balanced in an oven set at a relative humidity (44%, saturated solution of potassium carbonate), then were treated with the following products (Annex 2):

- placebo hydrogel (no active ingredient),
- hydrogel containing UFS dry spheres at 0.004% corresponding to 2% of UFS commercial product.

There was three consecutive applications spaced 30 minutes apart and applied doses were 1 mg/cm² (12 trials).

Assay method: dielectric conductivity measurement (Obata *et al*, 1990)

Dielectric conductivity (conductance) measurements were carried out prior to treatment and then up to 6 hours after the third application.

Results and statistics

The measurements were performed in triplicate on each sample and for each time point.

The moisturizing effect is expressed as a percentage of increase of the dielectric conductivity referring to the control without treatment and presented as a percentage of increase *vs* untreated control.

Statistical analysis was done using Student t test according to Sigmaplot software recommendation (Systat Software Inc, USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

★ New test - Volumizing efficacy on the lips

in vivo

OBJECTIVE

In a double-blind, placebo controlled and randomized clinical study, we evaluated the lip volumizing efficacy of UFS.
This study was performed on 33 female volunteers aged 45-65 years old and with Fitzpatrick skin type I, II, or III. At each study visit, a liquid lipstick formulation of either the placebo or UFS at 2% was applied to the lips (see annex 2). The volumizing efficacy of UFS and the placebo was evaluated using AEVA Imaging and Clinical Grading. Measurements were taken at timepoints corresponding to the baseline, 1 hour, and 2 hours.

RESULTS & DISCUSSION

After 1 hour, UFS at 2% showed a trend of increase in positive volume (p=0.09). After 2 hours, UFS 2% significantly increased by 4% the lip volume (Figure 15).

For the Clinical grade of superior lip volume, UFS at 2% showed a significant increase (p<0.001) of 30% and 36%, at 1 and 2 hours respectively, as compared to baseline (Figure 16).

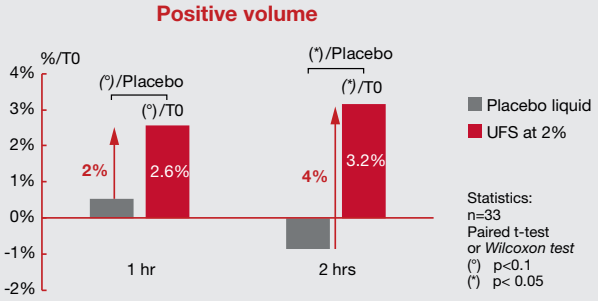


Figure 15 - Change in Positive Volume Over Time (AEVA).

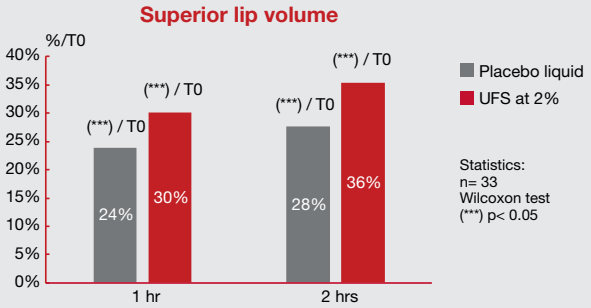


Figure 16 - Change in Superior Lip Volume Grade Over Time (Clinical Grading).

CONCLUSION

These results suggest that 2% UFS has volumizing efficacy for lips.

MATERIALS&METHODS

Study design

The clinical study was carried-out in randomized, double-blind, *vs* placebo way. The efficacy of the liquid lipstick formulation containing UFS at 2% was compared to the placebo. The formulation is detailed in Annex 2.

Inclusion criteria

The study was done on 33 female volunteers, aged 45-65 years old and with Fitzpatrick skin type I, II, or III.

Application modality

At each study visit, a liquid lipstick Formulation of either the placebo or UFS at 2% was applied to the lips. One product was applied to the lips per visit, over the course of two visits, as randomly assigned. Product application was overseen by a clinical scientist.

Evaluation methods

AEVA

The AEVA is an imaging system which uses stereo imaging to create three dimensional topographical images of the face. The system can then analyze these images to assess changes in facial topography, such as pore size, wrinkle depth, cheek curvature, lip volume, and skin roughness.

The software assesses the positive volume of topographical features of the face – such as the lip by analyzing the height of features above a certain threshold.

Clinical Grading

The overall condition of the subjects' lips was assessed by trained clinical scientists during the study visit. The subjects' lip volume was graded by the scientists at time points corresponding to the baseline, 1 hour, and 2 hours. The grading scale used for lip volume analysis was the lip volume grading scale (Rossi *et al*, 2011).

Statistics

All data has been analyzed to confirm the normality of the distribution using the Shapiro-Wilk Test. The following tests were used to compare the change in parameters of instrumental values:

- for those comparisons where the normality of both data sets was validated, a paired t test was used,
- if there was no validation of the normality, then the Wilcoxon Singed Rank Test was used. The Wilcoxon Test was used to compare the change in parameters of clinical grading values.

EFFICACY

★ New test - Smoothing efficacy on crow's feet wrinkles

in vivo

OBJECTIVE

In a double-blind, hemi-face and placebo controlled and randomized study, we evaluated the face smoothing efficacy of UFS. This study was performed on 27 female volunteers aged 45-65 years old and with Fitzpatrick skin type I, II, or III. At each study visit, an oily stick formulation of either the placebo or UFS at 2% was rubbed around the eye area.
The smoothing effect of UFS and the placebo on crow'feet wrinkles were evaluated using clinical grading at timepoints corresponding to the baseline, 1 hour and 5 hours.

RESULTS & DISCUSSION

UFS at 2% showed a significant decrease ($p<0.01$) of 10% in crow's feet wrinkle grade as compared to placebo at 5 hours. Additionally, UFS at 2% showed significant decreases ($p<0.001$ and $p<0.01$) in crow's feet wrinkle grade as compared to baseline at both 1 and 5 hours, of 11% and 13% respectively (Figure 17).

CONCLUSION

These results suggest that, unlike the placebo, UFS at 2% has a smoothing effect on crow's feet wrinkles that can last several hours.

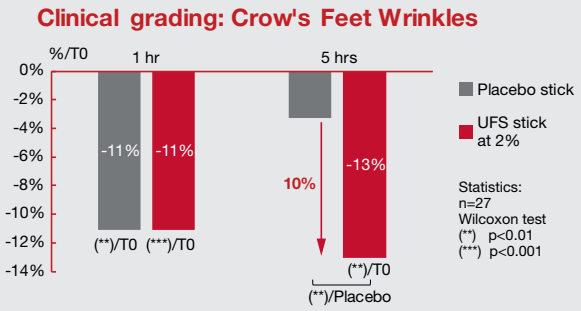


Figure 17 - Change in Crow's feet wrinkle grades.

MATERIALS&METHODS

Study design

The clinical study was carried-out in randomized, double-blind, vs placebo way. The efficacy of the oily stick formulation containing UFS at 2% was compared to the placebo. The formulation is detailed in Annex 2.

Inclusion criteria

The study was done on 27 female volunteers, aged 45-65 years old and with Fitzpatrick skin type I, II or III (see Annex 2 for the formulation composition).

Application modality

An oily stick of formulation of the placebo or UFS at 2% was rubbed around the eye area of either the left or right side of a clean and dry face, as randomly assigned. Product application was overseen by a clinical scientist.

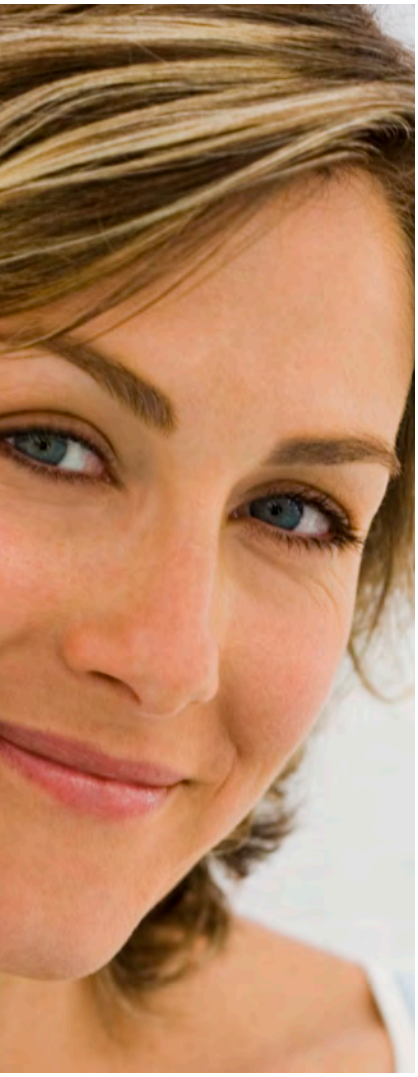
Evaluation methods

Clinical grading

The overall condition of the subjects'crow's feet wrinkles was assessed by trained clinical scientists during the study visit. The subject's crow's feet wrinkles were graded by the scientists at time points corresponding to the baseline, 1 hour, and 5 hours. The grading scale used for crow's feet analysis was the crow's feet wrinkle grading scale found in Bazin and Doublet's Skin Aging Atlas, Volume 1 (Bazin & Doublet, 2007).

Statistics

All data were analyzed to confirm the normality of the distribution using the Shapiro-Wilk Test followed by a Wilcoxon Singed Rank Test.



GENERAL CONCLUSION

Continuing down the path of Filling Spheres, the Ultra Filling Spheres (UFS) are positioned as the most effective generation in terms of wrinkle-filling care due to their remarkable ability to swell in contact with water.

The original association of konjac glucomannan with hyaluronic acid confers exceptional hygroscopic properties to UFS.

With their presence after application in the superficial layers of the epidermis, they allow water uptake by swelling even more rapidly. Through their physical action, they can help fill deep wrinkles and volumize the lips, without resorting to cosmetic surgery.

Subsequently, by retaining water longer, the UFS provide prolonged moisturizing comfort to better fight skin dryness.

This new generation has been developed to dim the appearance of lines and wrinkles and to protect the skin from harsh weather conditions, and especially to prevent dryness.

With this new generation, BASF Beauty Creations further increases the effectiveness of Filling Spheres. The efficacy tests presented above prove it: UFS capture more water, swell more and faster, and retain the water collected for a longer period of time than previous Filling Spheres.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and Regulatory Product Information
- Halal statement
- Natural content (according ISO 16128)
- Vegan statement
- Information on toxicological data
- Composition sheet
- Specifications
- Product Purity Profile

Annex 2 - Clinical test formula

Lipstick

Trade name	INCI name	Placebo formulation %	UFS fomulation %
Indopol H100 (INEOS)	Polybutene	44.00	44.00
Cosmedia® DC	Hydrogenated dimer dilinoleyl	5.00	5.00
Eutanol® G	Dimethylcarbonate copolymer	10.00	10.00
Cegesoft® C24	Octyldodecanol	12.00	10.00
Cetiol® Ultimate	Undecane, tridecane	8.00	8.00
Dowanol EpH (Dow)	Phenoxyethanol	1.00	1.00
Cosmedia® Gel CC	Dicaprylyl Carbonate, Stearalkonium, Hectorite, Propylene, Carbonate	5.00	5.00
Aerosil R974 (Evonik)	Silica Dimethyl Sylicate	5.00	5.00
DK-PGT PASTE R7 (Daito Kasei)	CI 15850, Polyglyceryl-2 Triisostearate	10.00	10.00
UFS C00487	Ethylhexyl Palmitate, Trihydroxystearin, Sodium hyaluronate, Glucomannan	-	2.00

Tagami test - Hydrogel

Trade name	INCI name	UFS fomulation %
Water	Water	88.296
Elestab 50J	Chlophenesin (and) Methylparaben	0.35
Water	Water	10.00
Rheocare® C Plus	Carbomer	0.60
NaOH 16% solution	Sodium Hydroxide, Water	0.75
UFS	Ethylhexyl Palmitate, Trihydroxystearin, Sodium hyaluronate, Glucomannan	0.004

Oily Face Stick

Trade name	INCI name	Placebo formulation %	UFS fomulation %
Cutina® HR Powder	Hydrogenated Castor Oil	9.00	44.00
Lanette® O	Cetearyl Alcohol	10.00	10.00
Cutina® GMS-V	Glyceryl Stearate	7.00	7.00
Cetiol® MM	Myrisryl Myristate	4.00	4.00
Myritol® 312	Caprylic/Capric Triglyceride	27.00	25.00
Kahlwax 2442 (Kahl&Co)	Camauba Wax	10.00	10.00
6001 Microwax (Khal&Co)	Ozokerite	6.00	6.00
Cetiol® RLF	Caprylyl Caprylate/Caprate	12.00	12.00
Cetiol® OE	Dicaprylyl Ether	5.00	5.00
Tinosorb® S	Bis-Ethylhexyloxyphenol Methoxyohenyl Triazine	4.00	4.00
Uvinul® MC 80	Ethylhexyl Methoxycinnamate	3.00	3.00
Uvinul® N359T	Octocrylene	3.00	3.00
UFS C00487	Ethylhexyl Palmitate, Trihydroxystearin, Sodium hyaluronate, Glucomannan	-	2.00

Annex 3 - Formulation examples

Youth serum (SC-RU-18-TK-03-01)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	65.50	
	Edeta® BD	Disodium EDTA	0.05	Complexing agent
	Glycerin	Glycerin	2.00	Humectant
	Rheocare® XGN	Xanthan Gum	0.20	Rheology modifier
	Pluracare® E 400	PEG-8	1.00	Humectant
B	Eumulgin® Prisma	Disodium Cetearyl Sulfosuccinate	1.00	Emulsifier (O/W)
	Cetiol® 4 All	Dipropylheptyl Carbonate	8.00	Emollient
	Cetiol® C 5	Coco-Caprylate	8.00	Emollient
	Cetiol® Ultimate	Undecane, Tridecane	5.00	Emollient
	Ultra Filling Spheres™ C00487	Ethylhexyl Palmitate, Trihydroxystearin, Sodium Hyaluronate, Glucomannan	2.00	Active ingredient
C	Tinovis® GTC UP	Acrylates/Beheneth-25 Methacrylate Copolymer	2.00	Rheology modifier
D	Replexium™ BC10091	Aqua, Dimethyl Isosorbide, Polysorbate 20, Acetyl Tetrapeptide-11, Acetyl Tetrapeptide-9	2.00	Active ingredient
	Dry-Flo PC *(Akzo Nobel)*	Aluminum Starch Octenylsuccinate	1.50	Skin feel modifier
	Euxyl PE 9010 *(Schülke)*	Phenoxyethanol, Ethylhexylglycerin	1.10	Preservative
	Perfume	Parfum	0.35	Fragrance
E	Sodium Hydroxide (25% solution)	Sodium Hydroxide	0.30	pH Adjustment

Natural instant blam to powder (SC-FR-20-009-A012)

Phase	Ingredients	INCI	% by weight	Function
A	Cutina® HR Flakes	Hydrogenated Castor Oil	2.00	Consistency agent
	Lanette® 22	Behenyl Alcohol	7.00	Consistency agent
	Cetiol® SB 45	Butyrospermum Parkii Butter	4.00	Consistency agent
	Eutanol® G	Octyldodecanol	13.30	Emollient
	Myritol® 331	Cocoglycerides	12.00	Emollient
	Cetiol® C 5C	Coco-Caprylate/Caprate	13.00	Emollient
	Cosmedia® Gel CC	Dicaprylyl Carbonate, Stearalkonium Hectorite, Propylene Carbonate	15.00	Rheology modifier
	Vivapur CS Sensory 15S *(JRS)*	Microcrystalline Cellulose, Cellulose Gum	15.00	Skin feel modifier
	Vivapur CS 9 FM *(JRS)*	Microcrystalline Cellulose	5.00	Skin feel modifier
B	Covi-ox® T 90 EU C	Tocopherol	0.20	Antioxidant
	Ultra Filling Spheres™ C00487	Ethylhexyl Palmitate, Trihydroxystearin, Sodium Hyaluronate, Glucomannan	3.00	Active ingredient
	Perfume	Parfum	0.50	Fragrance
	Cetiol® Ultimate	Undecane, Tridecane	10.00	Emollient

Lip balm - Cranberry fall (SC FR 20 BC 50902 02)

Phase	Ingredients	INCI	% by weight	Function
A	Lameform® TGI	Polyglyceryl-3 Diisostearate	6.00	Emollient
	Cosmedia® Gel CC	Dicaprylyl Carbonate, Stearalkonium Hectorite, Propylene Carbonate	3.00	Film forming agent
	Cosmedia® DC	Hydrogenated Dimer Dilinoleyl/Dimethylcarbonate Copolymer	1.35	Film forming agent
	Myritol® 331	Cocoglycerides	41.15	Emollient
	Cegesoft® C 24	Ethylhexyl Palmitate	11.00	Emollient
	Cetiol® CC	Dicaprylyl Carbonate	3.00	Emollient
	Cetiol® RLF	Caprylyl Caprylate/ Caprate	3.00	Emollient
	Sensiva SC 50 *(Schülke)*	Ethylhexylglycerin	1.00	Auxiliary
	Vitamin E-Acetate Care	Tocopheryl Acetate	1.00	Antioxidant
	Sunsil-130 *(Sunjin Chemical)*	Silica	1.00	Skin feel modifier
	Ultra Filling Spheres™ C00487	Ethylhexyl Palmitate, Trihydroxystearin, Sodium Hyaluronate, Glucomannan	3.00	Active ingredient
B	Candelilla Wax SP 75 *(Strahl & Pitsch)*	Candelilla Cera	6.00	Structurant
	Cutina® PES	Pentaerythrityl Distearate	6.00	Structurant
	Cegesoft® HF 62	Hydrogenated Vegetable Oil	6.00	Consistency agent
	Cetiol® SB 45	Butyrospermum Parkii Butter	4.00	Emollient
C	Chione™ Electric Sunset SK90D	Synthetic Fluorphlogopite, Titanium Dioxide, Iron Oxides, Silica	2.90	Effect pigment
	Cellini® Blue 620CB1C	Mica, Titanium Dioxide, FD&C Blue No. 1 Aluminum Lake, Hydrogenated Polyisobutene, Palmitic Acid	0.30	Effect pigment
	Timica® Terra Black MN4498	Mica, Iron Oxides, Titanium Dioxide	0.30	Effect pigment

Disclaimer

While the descriptions, designs, data and information contained herein are presented in good faith and believed to be accurate, it is provided gratis and for your guidance only. Because many factors may affect processing or application/use, we recommend that you make tests to determine the suitability of a product for your particular purpose prior to use. NO WARRANTIES OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE MADE REGARDING PRODUCTS DESCRIBED OR DESIGNS, OR THAT DATA OR INFORMATION MAY BE USED WITHOUT INFRINGING THE INTELLECTUAL PROPERTY RIGHTS OF OTHERS. IN NO CASE SHALL THE DESCRIPTIONS, INFORMATION, DATA OR DESIGNS PROVIDED BE CONSIDERED A PART OF OUR TERMS AND CONDITIONS OF SALE.
For more information visit us at www.personal-care.basf.com or www.carecreations.basf.com

BIBLIOGRAPHY

Al-Ghazzewi FH, Tester RF.
Effect of konjac glucomannan hydrolysates andprobiotics on the growth of the skin bacterium.
Propionibacterium acnes in vitro.
International Journal of Cosmetic Science, 32, 139–142, 2010

Alvarez-Manceñido F, Landin M, Lacik I, Martínez-Pacheco R.
Konjac glucomannan and konjac glucomannan/xanthan gum mixtures as excipients for controlled drug delivery systems. Diffusion of small drugs.
Int J Pharm. Feb 12;349(1-2):11-8, 2008

Arvill A, Bodin L.
Effect of short-term ingestion of konjac glucomannan on serum cholesterol in men.
Am J Clin Nutr 61:585-9, 1995

Bazin R, Doublet E.
Skin Aging Atlas.
Volume 1. Paris: MED'COM Publishing, p. 41, 2007

Chuaa M, Timothy C, Baldwina TC, Hockinga TJ, Chana K.
Traditional uses and potential health benefits of Amorphophallus konjac K. Koch ex N.E.Br.
J of Ethnopharmacology 128, 268–278, 2010

Heber G.
Composition and method for dermal regeneration.
Brevet WO2009/043111. 03.10, 2008

Kaneko C, Kobayashi TK, Kiyoshi H, Udagawa Y, Iwai M.
A Cell-Block Preparation Using Glucomannan. Extracted From Amorphophallus konjac.
Diagnostic Cytopathology, Vol 38, 9, 2010

Kligman AM, Papa CM.
Albumin as an antiwrinkling cosmetic.
J. Soc. Cosmetic Chemists, 16,557-562, 1965

Kondo T, Shinozaki T, Oku H, Takigami S, Takagishi K.
Konjac glucomannan-based hydrogel with hyaluronic acid as a candidate for a novel scaffold for chondrocyte culture.
J Tissue Eng Regen Med, 3: 361–367, 2009

Lu XJ, Chen XM, Fu DX, Cong W, Ouyang F.
Effect of Amorphophallus Konjac oligosaccharides on STZ-induced diabetes model of isolated islets.
Life Sci. Dec 27;72(6):711-9, 2002

Obata M, Tagami H.
Electrical determination of water content and concentration profile in simulation model of *in vivo stratum corneum.*
J. Invest. Dermatol., 92:854-859, 1989

Rochefort A, Drouot P, Leduc M, Vassalet R, Agache P.
A new technique for evaluation of cosmetics effect on mechanical properties of *stratum corneum* and epidermis *in vitro.*
Int. J. Cosm. Sci., 8:27-36, 1986

Rossi AB.
Development and Validation of a Photonumeric Grading Scale for Assessing Lip Volume and Thickness.
J Eur Acad Dermatol Venerol, 25, 523-531, 2011

Vasques Carlos AR, Rossetto S, Halmenschlager G, Linden R, Heckler E, Poblador Fernandez Maria S, Lancho Alonso José L.
Evaluation of the Pharmacotherapeutic Efficacy of Garcinia cambogia plus *Amorphophallus konjac* for the Treatment of Obesity.
Phytother, Res. 22, 1135–1140, 2008

Europe

BASF Beauty Creations

49, avenue Georges Pompidou
92593 Levallois-Perret Cedex
France

Phone: +33 (0) 1 49 64 52 09
bcs-europe@basf.com

North America

BASF Corporation
North American Regional Headquarters
100 Park Avenue
Florham Park, NJ 07932
USA

Phone: +1-800-962-7831
CosmeticsCustomerCare@basf.com

South America

BASF S.A.
Av. das Nações Unidas, 14171
Crystal Tower
04794-000 - São Paulo - SP
Brazil

Phone: +55 11 2039 2273
personal-care-sa@basf.com

Asia Pacific

BASF East Asia
Personal Care
45/F Jardine House
No.1 Connaught Place
Central, Hong Kong

Phone: +852 2731 0190
personal-care-hk@basf.com

Publisher
BASF Beauty Care Solutions France SAS
3, Rue de Seichamps - 54425 Pulnoy (France)
Email: bcs-europe@basf.com
www.carecreations.basf.com

Pictures
© BASF Beauty Care Solutions France SAS. All rights reserved.
®/™ Registered trademark / trademark of BASF in several countries

Global Edition, May 2013 (update May 2022)

Although all statements and information in this publication are believed to be accurate and reliable, they are presented gratis and for guidance only, and risks and liability for results obtained by use of the products or application of the suggestions described are assumed by the user. SELLER MAKES NO WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, BY FACT OR LAW, INCLUDING WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Statements or suggestions concerning possible use of the products are made without representation or warranty that any such use is free of patent infringement and are not recommendations to infringe any patent. The user should not assume that toxicity data and safety measures are indicated or that other measures may not be required. The claims and supporting data provided in this publication have not been evaluated for compliance with any jurisdiction's regulatory requirements and the results reported may not be generally true under other conditions or in other matrices. Users must evaluate what claims and information are appropriate and comply with a jurisdiction's regulatory requirements. Recipient of this publication agrees to (1) indemnify and hold harmless each entity of the BASF organization for any and all regulatory action arising from recipient's use of any claims or information in this publication, including, but not limited to, use in advertising and finished product label claims, and (2) not present this publication as evidence of finished product claim substantiation to any regulatory authority.